



Simple Ways for Swimming Pool Leak Detection

Understanding Water Loss

Water loss in a swimming pool can often be mistaken for **evaporation or humidity**, which is normal. In a moderate climate, a pool can naturally lose up to **2 inches of water per week**. However, if you observe your pool's water level dropping by **more than 2 inches** weekly, it's a strong indicator that you have a leak that needs investigation. Over time and with regular use, the pool's initial water tight sealants wear out, and parts of the structure may suffer erosion, leading to leaks. These leaks can originate from problems in the pool's fittings, accessories, or due to wear and tear in the plumbing system.

Confirming a Leak :The Bucket Test

Before conducting extensive tests, you should confirm that a leak actually exists. A simple and effective method is the "**Bucket Test**." Place a bucket of water on the swimming pool steps, ensuring the top of the bucket is positioned **just below the pool's water level**. Leave the bucket for a few days and then carefully compare the water level inside the bucket to the pool's water level outside the bucket. If the pool's water level has dropped significantly more

Than the water level in the bucket (which accounts for environmental loss like evaporation), this confirms that your pool has a leak, and the next step is to locate the source.

Locating the Leak: The Filter System Comparison

Once a leak is confirmed, you can determine if the problem lies with the plumbing or the structure by comparing water loss with the filter system running and off. First, **fill the pool to its normal level** and mark it with duct tape. Run the filter system for 12-24 hours and measure the water loss. Then, **refill the pool** to the normal level, switch the **filter system off**, and measure the water loss over the same duration, ideally at the same time of day.

Comparing the measurements:

- **More loss with the filter running:** The leak is in the **pressure side** of the plumbing (after the pump's impeller).
 - **Less loss with the filter off:** The leak is in the **vacuum side** of the plumbing (before the impeller).
 - **Same loss in both cases:** The issue is with the **pool's structure**, not the plumbing.
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Finding Structural Cracks and Professional Help

If the leak is not found in the visible plumbing, or if the water loss is the same with the filter on and off, it suggests a problem with the **pool structure** or a hidden plumbing issue. To check for structural leaks, **turn off the pump for an hour**. Then, look for any visible cracks on the pool walls. Squeeze a few drops of **food colouring** near a suspected crack. If the colour is immediately **sucked into the crack**, you've found a leak source. This be patched and re-tested.

For leaks that remain hidden or are difficult to access, especially with in the plumbing, it is often best to **call in professionals**. Experts use specialized equipment like **geophones** to precisely locate the leak underground or within the pool structure. This focused detection prevents unnecessary digging, saving you significant time and effort compared to pressure testing and manually digging up lines yourself.
